

# WOOD CUTTING PROCESS REPORT

Process #1021

## PROCESS INFORMATION

|            |                     |
|------------|---------------------|
| Process ID | 1021                |
| Start Time | 28/08/2026 17:29:23 |
| End Time   | 28/08/2026 17:30:04 |
| Duration   | 00:00:40            |
| Wood Type  | Lignamon 783        |
| Category   | Hardwood            |

## PRODUCTION SUMMARY

|                              |
|------------------------------|
| Product Width : 30 mm        |
| Product Height : 200 mm      |
| Production Volume : 0.000 m³ |
| Total Cost : 0.12            |

## MACHINE CONDITIONS

|                      |               |
|----------------------|---------------|
| Cutting Speed        | 3840.00 m/min |
| Feed Rate            | 11.00 m/min   |
| Shaft Speed          | 3492 RPM      |
| Consumed Electricity | 0.00 kWh      |

## CRITICAL CUTTING ANALYSIS

|                       |           |
|-----------------------|-----------|
| Cutting Force         | 82.61 N   |
| Active Force          | 83.75 N   |
| Friction Force        | 46.91 N   |
| Thrust Force          | 13.77 N   |
| Shear Force           | 56.17 N   |
| Normal Force To Shear | 60.58 N   |
| Normal Force To Rake  | 67.99 N   |
| Cutting Moment        | 14.46 N.m |

## PERFORMANCE INDICATORS

|                      |                                |                                   |
|----------------------|--------------------------------|-----------------------------------|
| Duration<br>00:00:40 | Production Rate<br>0.009 m³/hr | Energy Efficiency<br>0.059 m³/kWh |
|----------------------|--------------------------------|-----------------------------------|